

Sophia Chan Davis

Curriculum Vitae

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EDUCATION

University of Texas at San Antonio

August 2025 – Present

PhD in Space Physics

Graduate Course Highlights: Fundamentals of Space Physics

Graduate Relevant Coursework: Classical Mechanics I, Electrodynamics I

University of Michigan, Ann Arbor

August 2021 – May 2025

Bachelor of Science in Astronomy & Astrophysics and Interdisciplinary Physics

Minor in Museum Studies

Undergraduate Course Highlights:

Modern Astronomical Techniques, Galaxies & the Universe, Stellar Astrophysics, Scientific Communication, Scientific Writing, High Energy Astrophysics, Exoplanets

Undergraduate Relevant Coursework:

Houghton Scholars Calculus Workshop I & II, Differential Equations, Methods of Theoretical Physics I, Intermediate Mechanics, Intermediate Electromagnetism, Museums & Society, Museum Work & Critical Practice

Undergraduate Organizations:

Douglass Houghton Scholars Program, Meijer Scholars Program, Community of Physicists for Equity and Inclusion, Student Astronomical Society, Mixed at Michigan, Collaborative Lab for Advancing Work in Space

RESEARCH EXPERIENCE

Astronomy Research Assistant

May 2024 – May 2025

Maria Mitchell Association

Astronomy REU Program

Investigated the origins of planetary nebulae in globular clusters using advanced X-ray observations under the guidance of Dr. Rodolfo Montez Jr. and Dr. Regina Jorgenson. Analyzed two decades of archival data from the Chandra X-ray Observatory using Python-based tools and MARX simulation software to process and interpret X-ray emission.

Solar Physics Research Assistant

May 2023 – May 2024

Harvard-Smithsonian Center for Astrophysics

Solar and Stellar X-ray Group REU Program

Assessed the accuracy of measuring interplanetary shockwave impacts on Earth using data from four spacecraft (Wind, ACE, DSCOVR, SOHO) under the mentorship of Dr. Michael Stevens, Lucas Guliano, and Dr. Tatiana Niembro Hernandez. Conducted statistical and geometric analyses, identified shock candidates, and performed Rankine-Hugoniot (R-H) diagnostics.

Research Co-Lead

September 2022 – May 2025

University of Michigan

Collaborative Lab for Advancing Work in Space (CLAWS)

Led research efforts for the NASA SUITS challenge, collaborating with a multidisciplinary team to develop an Augmented Reality (AR) interface supporting astronaut tasks during lunar and Martian missions. Researched AR technologies and Extravehicular Activities (EVAs), co-authored academic papers, and led research meetings to guide design and development.

Astronomy Research Assistant

May 2022 – May 2024

University of Michigan

Worked under PhD candidate Anne Blackwell and Dr. Joel Bregman on high-energy extragalactic spectroscopy, addressing the Missing Metal Conundrum in galaxy clusters. Performed X-ray data reduction and spectral analysis using data from Chandra and XMM-Newton. Utilized tools including Bash, SAS, Xspec, SAOImage DS9, CIAO, and XQuartz for calibration and interpretation.

Mathematics Research Assistant

October 2021 – July 2022

University of Michigan

Undergraduate Research Opportunity Program

Collaborated on mathematical research with Prof. Steven J. Miller (Williams College) to generalize depth-two recurrence sequences and the Tribonacci numbers. Co-authored “Reciprocal Sums of Recurrence Relations,” published in the *Fibonacci Quarterly* (Dec 2022). Used Mathematica for advanced calculations and presented findings at the 20th International Fibonacci Conference.

EMPLOYMENT / RELEVANT EXPERIENCE

Beta Tester – Space Umbrella Citizen Science Platform July 2025
Zooniverse

Tested and provided feedback on Space Umbrella, a Zooniverse-based platform developed to identify magnetopause crossings using data from NASA's Magnetospheric Multiscale (MMS) mission. Evaluated classification flow, field guide and tutorial content, and interface usability. Contributed to the improvement of public-facing heliophysics research tools.

STEM Cohort Member August 2024 – June 2025
Alliance for the Low-Income & First-Generation Narrative (AL1GN)

Participated in a national community supporting first-generation and low-income students in STEM. Attended monthly workshops and weekly stand-ups focused on mentorship, identity-building, and professional development. Concluded the year at the AL1GN Conference in NYC, hosted by the Flatiron Institute.

Astronomy Educator June 2024 – August 2024
Maria Mitchell Association

Designed and delivered engaging astronomy education programs for the Junior Astronomers Camp (ages 7–11). Created weekly lesson plans aligned with camp goals and educational standards. Adapted activities based on counselor feedback to foster excitement and understanding of astronomy through hands-on learning.

Astronomy Outreach Liaison May 2024 – August 2024
Maria Mitchell Association

Led twice-weekly public open nights, delivering talks on the observatory's history and guiding constellation tours. Operated both an 8-inch manual and 24-inch computerized telescope, offering context on celestial objects and telescope functionality. Helped broaden public engagement with astronomy.

Science Communication Consultant – Watershed Education April 2024
University of Michigan & Huron River Watershed Council

Collaborated with HRWC through a course project in ALA 270: Science Communication to evaluate the accessibility of their educational materials. Conducted user-testing interviews with K–12 educators and proposed website redesign strategies to improve usability and educator autonomy.

COFFIES Team Member February 2024 – Present
Stanford University – NASA DRIVE Science Center

Contribute to a heliophysics center focused on developing data-driven models of solar activity. Support early-career engagement through the Beans community and participate in the POISE Initiative to enhance public enthusiasm for NASA science. Assist with outreach efforts and facilitate student engagement with COFFIES programs.

Planetarium Presenter September 2022 – May 2025
University of Michigan Museum of Natural History

Delivered immersive planetarium shows tailored for diverse audiences. Operated the Digistar system and managed lighting, sound, and startup/shutdown procedures. Created and shared 3D visual content for the Digistar cloud, advancing public understanding of astrophysical concepts.

Undergraduate DEI Representative October 2022 – May 2025
University of Michigan Department of Astronomy

Served on departmental committees to address issues of diversity, equity, and inclusion within the undergraduate astronomy community. Acted as a liaison between students and faculty, advocating for a more inclusive academic environment through collaborative initiatives.

NASA Solar Eclipse Ambassador Partner October 2022 – April 2024
NASA & Astronomical Society of the Pacific

Partnered with Buddy Stark to conduct educational outreach events across libraries and museums in metro Detroit, reaching over 250 K–12 students and community members. Promoted safe solar viewing and eclipse science during the 2023 and 2024 eclipses. Trained in public engagement by NASA and ASP; received local media recognition.

PUBLICATIONS

Academic Papers

Illuminating the Augmented Reality Stage: Assessment of Lighting Conditions for Space Exploration

N. Alesawy, P. Halim, M. Nadai, J. Son, **S. Davis**, C. Teixeira, T. Nelson, A. Wang, G. Holliman-Lopez, S.

Alesawy; advised by L. Stirling

Accepted to Human Factors and Ergonomics Society 67th International Annual Meeting

March 2023

Sums of Reciprocals of Recurrence Relations

H. Cui, X. Cui, **S. Davis**, I. Durmic, Q. Hu, L. Liu, S. J. Miller, F. Ren, A. S. Reina, E. Sosis

Fibonacci Quarterly, Volume 60, Issue 5

December 2022

PRESENTATIONS / CONFERENCE ACTIVITY

Shining the X-ray Light on Planetary Nebulae in Globular Clusters: K648 in M15

Poster — Undergraduate Research Symposium / University of Michigan / April 2025

With R. Montez

Shining the X-ray Light on Planetary Nebulae in Globular Clusters: K648 in M15

Public Talk — Society of Physics Students / University of Texas at San Antonio / March 2025

With R. Montez

Shining the X-ray Light on Planetary Nebulae in Globular Clusters: K648 in M15

Poster — 245th Meeting of the American Astronomical Society / January 2025

With R. Montez

Shining the X-ray Light on Planetary Nebulae in Globular Clusters

Public Talk — Student Astronomical Society Research Series / University of Michigan / October 2024

With R. Montez, R. Jorgenson

Shining the X-ray Light on Planetary Nebulae in Globular Clusters

Public Talk — Maria Mitchell Association Science Speaker Series / August 2024

With R. Montez, R. Jorgenson

Contrast Sensitivity and Visual Acuity in Red-Lit Environments: A Mixed Reality Study for Mars Exploration

Poster — NASA Exploration Science Forum / University of Michigan / July 2024

With T. Nelson, K. Chandra, K. Joshi, K. Kumar, G. Holliman-Lopez, M. Hart, A. Rajesh

Shock Waves in Space: Can One Satellite Tell Us What's Coming, or Do We Need a Fleet?

Public Talk — Michigan Astronomy Undergraduate Research Symposium / University of Michigan / April 2024

With L. Guliano, T. N. Hernández, M. L. Stevens

Shock Waves in Space: Can One Satellite Tell Us What's Coming, or Do We Need a Fleet?

Poster — Conference for Undergraduate Women in Physics / University of Michigan / January 2024

With L. Guliano, T. N. Hernández, M. L. Stevens

Shock Waves in Space: Can One Satellite Tell Us What's Coming, or Do We Need a Fleet?

Poster — American Geophysical Union (AGU) Fall Meeting / December 2023

With L. Guliano, T. N. Hernández, M. L. Stevens

Shock Waves in Space: Can One Satellite Tell Us What's Coming, or Do We Need a Fleet?

Public Talk — Student Astronomical Society Research Series / University of Michigan / September 2023

With L. Guliano, T. N. Hernández, M. L. Stevens

Shock Waves in Space: Can One Satellite Tell Us What's Coming, or Do We Need a Fleet?

Public Talk — CfA Solar REU Symposium / Harvard-Smithsonian Center for Astrophysics / August 2023

With L. Guliano, T. N. Hernández, M. L. Stevens

Innovation with Emerging Technologies: Collaborative Lab for Advancing Work in Space at the University of Michigan

Poster — NASA SUITS Poster Session / Johnson Space Center / May 2023

With N. Alesawy, P. Halim, M. Nadai, C. Teixeira, T. Nelson, A. Wang, G. Holliman-Lopez, S. Alesawy; advised by L. Stirling

Missing Metal Conundrum: Measuring Metallicity Content in Galaxy Clusters

Poster — Michigan Astronomy Undergraduate Research Symposium / University of Michigan / April 2023

With A. Blackwell; advised by J. Bregman

Augmented Reality in Space and in Surgery: Assessing HoloLens 2's Compatibility with Lighting and Hand Coverings

Poster — Global Surgical Innovation National Conference / University of Michigan / April 2023

With N. Alesawy, P. Halim, M. Nadai, C. Teixeira, T. Nelson, A. Wang, G. Holliman-Lopez, S. Alesawy, T. Kantor, R. Reddy; advised by L. Stirling

Missing Metal Conundrum: Measuring Metallicity Content in Galaxy Clusters

Poster — Celebrating Donors & Scholars Luncheon / University of Michigan / March 2023

With A. Blackwell; advised by J. Bregman

Sums of Reciprocals of Recurrence Relations

Public Talk — International Fibonacci Conference / University of Sarajevo / July 2022

With I. Durmic, A. S. Reina, E. Sosis; advised by S. J. Miller

Sums of Reciprocals of Recurrence Relations

Poster — UROP Spring Research Symposium / University of Michigan / April 2022

With E. Sosis; advised by S. J. Miller

OUTREACH AND SCIENCE COMMUNICATION

Community Building & Science Engagement

Femmes in Astrophysics

UMMNH with Women in Science and Engineering (WISE), March 2025

Led interactive planetarium presentations for 120 WISE Residential Program students. Presented visible constellations, planetary motion, and astronomical phenomena while incorporating my own research, career story, and the importance of accessible science communication. Created space for honest conversation around identity and representation in astrophysics.

Saturday Morning Physics: Good Vibrations

University of Michigan, March 2025

Served as Stage & Exhibit Presenter. Delivered a live stage show on sound spectroscopy and led a hands-on optical spectroscopy exhibit that engaged families in decoding auroral spectra. Helped launch U-M's STEMobile with interactive physics-based programming.

Elemental Fingerprints of the Aurora

Scientist Spotlight Series, UMMNH, January 2025

Designed and led an interactive booth using spectroscopes and aurora emission spectra. Created an emission line matching game and answered visitor questions on auroras, space weather, and solar particles.

NASA Eclipse Ambassador Outreach Events (2022–2024)

As a NASA Eclipse Ambassador, led over five solar eclipse outreach events across southeast Michigan focused on underserved communities.

- “*All About Eclipses*”, Ann Arbor District Library — Developed tabletop activities; distributed eclipse glasses; reached 80+ attendees.
- “*Eclipse STEM Stations*”, Ypsilanti District Library, Whittaker — Engaged families in eclipse science through interactive games.
- “*Cosmic Coincidence*”, Scientist Spotlight Series at UMMNH — Created stations including “Making Pin-hole Projectors”; event drew over 600 attendees.
- “*Eclipse Ambassadors with the University of Michigan*”, Ypsilanti District Library, Superior — Facilitated eclipse activities and live Q&A to close the program.

Cambridge Science Festival: Cambridge Explores the Universe

Harvard-Smithsonian Center for Astrophysics, October 2023

Led public tours of the Solar Dynamics Observatory Mission Operations Center. Explained solar observatory instruments and solar physics to visiting families and students.

Features, Media, and Guest Appearances

NASA Aurorasaurus & Science Editorial

- “*Poetic Wisdom From Winter Field School*”, May 2025 — Featured for reflections on persistence, self-worth, and the scientific process.
- “*Warmth and Awe: Winter Field School Students on Seeing the Aurora*”, March 2025 — Authored piece blending personal aurora experience with heliophysics.
- “*Eclipses to Auroras*”, NASA Science Editorial, February 2025 — Profiled for participation in NASA's Winter Field School.

- “*Eclipses, Auroras, and the Spark of Becoming*”, NASA Science Editorial, May 2025 — Highlighted the field school’s influence on my early science identity.

Epiphany Essay Presentation

Eclipses to Aurora Winter Field School, University of Alaska Fairbanks, January 2025

Presented a reflective essay weaving together scientific learning, cultural insight, and aurora experiences to peers, mentors, and NASA educators.

COFFIES Newsletter

Consequence Of Fields and Flows in the Interior and Exterior of the Sun, April 2024

Featured for NASA Eclipse Ambassador outreach impact in heliophysics communication.

“Draco and the Naga Connection”

Maria Mitchell Association Podcast, August 2024

Invited guest speaker on cultural storytelling in astronomy, sharing the Cambodian Naga’s connection to the constellation Draco.

Student MC, Donors and Scholars Luncheon

University of Michigan, April 2024

Spoke as a Meijer Scholar about the transformative impact of scholarship support, fostering connections with donors and scholars.

Alumni Interview Panelist

Douglass Houghton Scholars Program, University of Michigan, March 2024

Participated in a panel guiding prospective scholars through academic and personal development experiences in STEM.

Press Coverage

WWJ Newsradio – “NASA Eclipse Ambassadors”, April 2024

Interviewed about my community outreach work and public engagement leading up to the 2024 solar eclipse.

Detroit Free Press – “Eerie Shadows, Quiet Wildlife”, by Matthew Dolan, April 2024

Highlighted during totality viewing with U-M Student Astronomical Society. Interviewed on eclipse science and reactions.

Bridge Michigan, April 2024

- “*Eclipse Blocks Sun, Brings Out Smiles from Michigan to Ohio*”, by Mike Wilkinson
- “*U-M Student on Mission as NASA Eclipse Ambassador*”, by Janelle D. James

Recognized for science outreach and community education as a NASA Ambassador.

Voice of America – Khmer – “Angkor Complex”, March 2024

Featured in a Khmer-language interview on the cultural significance of the “Angkor Complex” exhibit at UMMA.

The Michigan Daily – “Angkor Complex” Exhibit Coverage, by Astrid Code, February 2024

Quoted in coverage of the exhibit’s personal and community resonance, reflecting on Cambodian-American identity and museum representation.

AWARDS AND HONORS

Chandra/CIAO X-ray Data Workshop Participant

UMass Lowell / Chandra X-ray Center, May 2025

Selected for competitive workshop on advanced X-ray data analysis using the CIAO software suite. Trained with new tools such as `fine_astro` for astrometric merging on personal Chandra dataset of planetary nebula K648. Awarded \$3,000 travel grant.

Graduate Program Admissions – PhD Track in Space Physics

George Mason University, University of Texas at San Antonio, Georgia State University, April 2025

Received multiple fully funded offers to competitive doctoral programs based on research background, academic preparation, and faculty recommendations.

Michigan Difference Student Leadership Award

University of Michigan, March 2025

University-wide honor awarded to student leaders making meaningful contributions to campus, Michigan, and global communities through service, scholarship, and leadership.

NASA Winter Field School Participant

Astronomical Society of the Pacific / University of Alaska Fairbanks, January 2025

Selected among nine students nationally to attend this immersive program focused on heliophysics, auroras, and Indigenous star knowledge. Participated in lectures, site visits (e.g., Poker Flat Research Range), citizen science through Aurorasaurus, and science communication training. Awarded \$1,000 stipend.

FUTURE of Physics Workshop Participant

California Institute of Technology, September 2024

Selected for program supporting underrepresented students in physics and astronomy. Explored graduate life and research culture through mentoring and community building.

NASA Eclipse Ambassador Honorarium

Astronomical Society of the Pacific, July 2024

Awarded \$200 for leading public solar eclipse outreach events across southeast Michigan.

Outstanding Poster Presentation Award

Conference for Undergraduate Women in Physics (CUWiP), January 2024

Recognized for excellence in scientific presentation at the American Physical Society's regional CUWiP conference.

Travel Grant – American Geophysical Union Conference

Harvard-Smithsonian Center for Astrophysics, December 2023

Awarded \$2,500 to present research at AGU 2023. Covered travel, lodging, and per diem.

Travel Grant – Cambridge Science Festival

Harvard-Smithsonian Center for Astrophysics, October 2023

Awarded \$2,000 to support travel and participation in CfA's solar public engagement event.

NASA SUITS Challenge Finalist (Top 10)

NASA, 2023–2025

Recognized as one of ten top university teams nationally in the NASA SUITS (Spacesuit User Interface Technologies for Students) design challenge three years in a row.

Conference Travel Grant – International Fibonacci Conference

Williams College, July 2022

Awarded \$2,000 to attend and present number theory research internationally.

UROP Travel Grant

University of Michigan, July 2022

Received \$800 to support participation in the International Fibonacci Conference through the Undergraduate Research Opportunity Program.

Meijer Family Scholar

University of Michigan, 2021–2025

Awarded \$10,000 annually in support of academic excellence and community engagement.

Douglass Houghton Scholar

University of Michigan, 2021–2025

Selected for academic and leadership potential in STEM. Participated in a community of scholars from under-represented backgrounds.